Exponential Functions and Their Graphs

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating and describing

Let $f(x) = 3 \cdot 2^x$. Evaluate:

a $f(0)$ **b** $f(3)$ **c** $f(-2)$

2 [Foundational] • Target: Evaluating and describing

For $y = 5^x$, state:

a the domain

b the range

c the horizontal asymptote

d the y-intercept

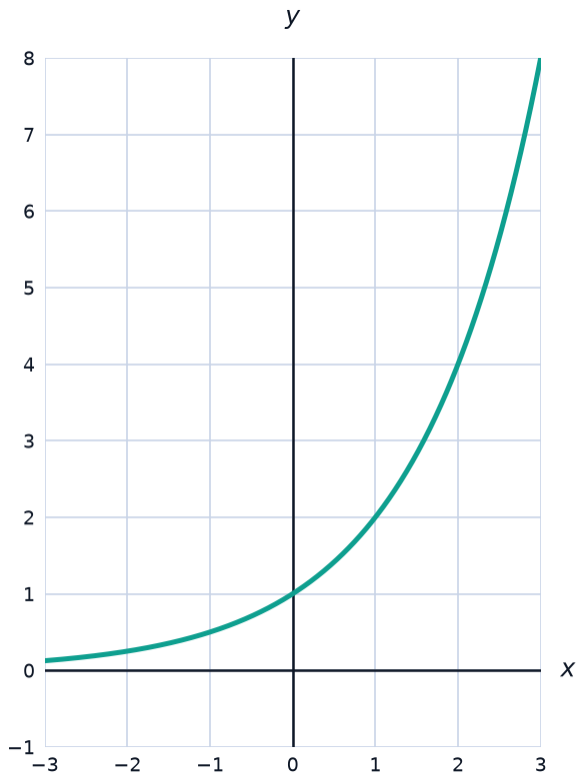
3 [Foundational] • Target: Evaluating and describing

Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of g.

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4 [Foundational] • Target: Identifying from graphs and applications

The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.



5 [Foundational] • Target: Identifying from graphs and applications

An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%: $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest riyal.

6 [Foundational] • Target: Identifying from graphs and applications

Determine whether each function represents growth or decay:

a $y = 4(1.08)^x$

b $y = 7(0.85)^x$

c $y = 2e^{-0.3x}$

d $y = \left(\frac{5}{4}\right)^x$

7 [Foundational] • Target: Compound interest

2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P(1 + \frac{r}{n})^{nt}$.

a Find the value after 3 years, to the nearest riyal.

b Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest riyal.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Compound interest

How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

9 [Intermediate] • Target: Comparing exponential growth rates

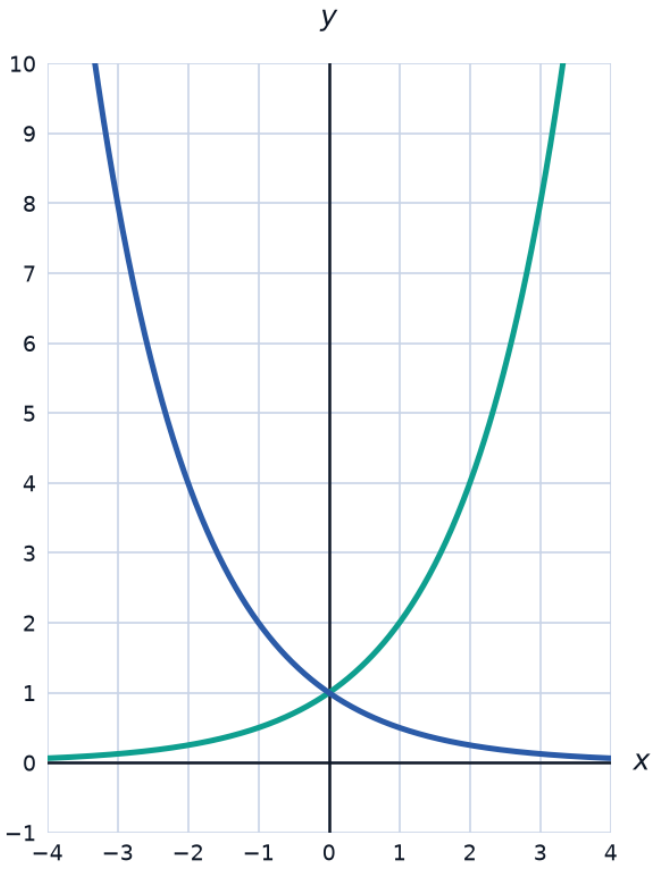
Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.

10 [Intermediate] • Target: Comparing exponential growth rates

Rank the following functions from slowest to fastest growth as $x \to \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.

11 [Intermediate] • Target: Comparing growth and decay graphically

The graphs of $y = 2^x$ (growth) and $y = \left(\frac{1}{2}\right)^x$ (decay) are shown together. Use the graph to describe the symmetry relating the two curves.



12 [Intermediate] • Target: The natural exponential base e

Explain what distinguishes $e \approx 2.71828$ from other exponential bases, in terms of the slope of $y = e^x$ at $x = 0$.

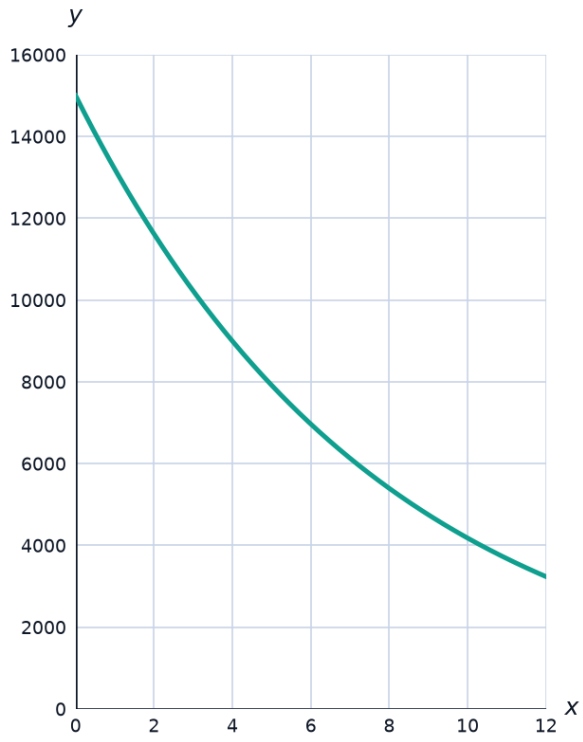
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13 [Intermediate] • Target: The natural exponential base e

Evaluate e^0, e^1 (to 3 decimal places), and simplify $e^{\ln 5}$.

14 [Intermediate] • Target: Piecewise-exponential applications

A machine depreciates in value according to $V(t) = 15000(0.88)^t$ (riyals, years). Find its value after 5 years, and the year in which its value first drops below 5000 riyals (to the nearest year).



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

State the horizontal asymptote and range of each transformed exponential function:

a $y = 3^x + 4$

b $y = -2^x$

c $y = 5^x - 3$

16 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Write the equation of the function obtained by taking $y = e^x$, reflecting it across the y-axis, then shifting it down 2 units.

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17 [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.01 The Unit Circle and Exact Values

For $f(x) = 3^x$, complete a short table of values at $x = -1, 0, 1, 2, 3$ and verify each value satisfies $f(x+1) = 3f(x)$.

18 [Challenging] • Target: Comparing growth and decay graphically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the inverse of $f(x) = \frac{4x-1}{2x+3}$.

19 [Challenging] • Target: The natural exponential base e • Cross-Lesson: 3.03 Sum and Difference Formulas

Solve $7^x = 100$ for x, expressed using logarithms.

20 [Challenging] • Target: Piecewise-exponential applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Base e and Exponential Growth/Decay Curves

For $f(x) = a \cdot b^x$ with $a > 0$, under what condition does $f(x)$ represent exponential growth?

a. $b > 1$

b. $0 < b < 1$

c. $b < 0$

d. $b = 1$

22 [Foundational] • Target: Base e and Exponential Growth/Decay Curves

Find the horizontal asymptote and y-intercept of $f(x) = 3 \cdot 2^x - 5$.

a. Asymptote: $y = -5$, y-intercept: $(0, -2)$

b. Asymptote: $y = 0$, y-intercept: $(0, 3)$

c. Asymptote: $y = -5$, y-intercept: $(0, -5)$

d. Asymptote: $x = 0$, y-intercept: $(0, -2)$

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23 [Foundational] • Target: Base e and Exponential Growth/Decay Curves

Determine the range of $g(x) = -2 \cdot e^x + 4$.

a. $(-\infty, 4)$

b. $[4, \infty)$

c. $(4, \infty)$

d. $(-\infty, -2)$

24 [Foundational] • Target: Transformations of Exponential Functions

Describe the transformations needed to obtain $g(x) = 2^{x-3} + 1$ from $f(x) = 2^x$.

a. Shift right by 3 units and up by 1 unit

b. Shift left by 3 units and up by 1 unit

c. Shift right by 3 units and down by 1 unit

d. Vertical stretch by 3 and up by 1

25 [Foundational] • Target: Transformations of Exponential Functions

What is the domain of any function of the form $f(x) = a \cdot b^{x-h} + k$ $(b > 0, b \neq 1)$?

a. $(-\infty, \infty)$

b. $[0, \infty)$

c. (k, ∞)

d. $[h, \infty)$

26 [Foundational] • Target: Transformations of Exponential Functions

Which transformation reflects the graph of $f(x) = e^x$ across the y-axis?

a. $g(x) = e^{-x}$

b. $g(x) = -e^x$

c. $g(x) = -e^{-x}$

d. $g(x) = e^x - 1$

27 [Foundational] • Target: Compound Interest Formulas

State the formula for an initial principal P compounded continuously at annual rate r for t years.

a. $A = Pe^{rt}$

b. $A = P(1 + r)^t$

c. $A = P(1 + \frac{r}{n})^{nt}$

d. $A = Prt$

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28 [Foundational] • Target: Compound Interest Formulas

If 1000 dollars is invested at 6% compounded monthly, what is the balance after 5 years?

a. $1000\left(1+\frac{0.06}{12}\right)^{60} \approx 1348.85$ dollars

b. 1300 dollars

c. 1349.86 dollars

d. 1338.23 dollars

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Compound Interest Formulas

If 2000 dollars is invested at 5% compounded continuously, find the value after 10 years.

a. $2000e^{0.5} \approx 3297.44$ dollars

b. 3000 dollars

c. 3257.79 dollars

d. 4000 dollars

30 [Intermediate] • Target: The Number e as a Limit

State the limit definition of Euler's number e.

a. $e = \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n$

b. $e = \lim_{n\to 0} \left(1+\frac{1}{n}\right)^n$

c. $e = \lim_{n\to\infty} (1+n)^{1/n}$

d. $e = \sum_{n=1}^{\infty} \frac{1}{n}$

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31 [Intermediate] • Target: The Number e as a Limit

Evaluate $\lim_{n\to\infty}\left(1+\frac{3}{n}\right)^n$.

a. e^3

b. $3e$

c. $e^{1/3}$

d. 3

32 [Intermediate] • Target: The Number e as a Limit

Why is continuous compounding optimal compared to discrete compounding at the same nominal rate?

a. As $n \to \infty$, the compounding factor $(1 + r/n)^{nt} \to e^{rt}$, which achieves the theoretical maximum return

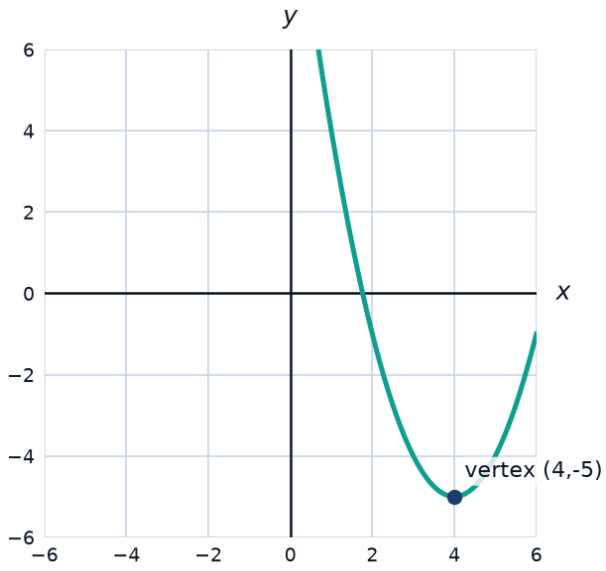
b. It avoids paying bank fees

c. The interest rate increases

d. It doubles every year

33 [Intermediate] • Target: The Number e as a Limit

The graph of $g(x) = f(x - 4) - 5$ is obtained from $f(x)$ by:



a. Shifting 4 units left and 5 units up

b. Shifting 4 units right and 5 units down

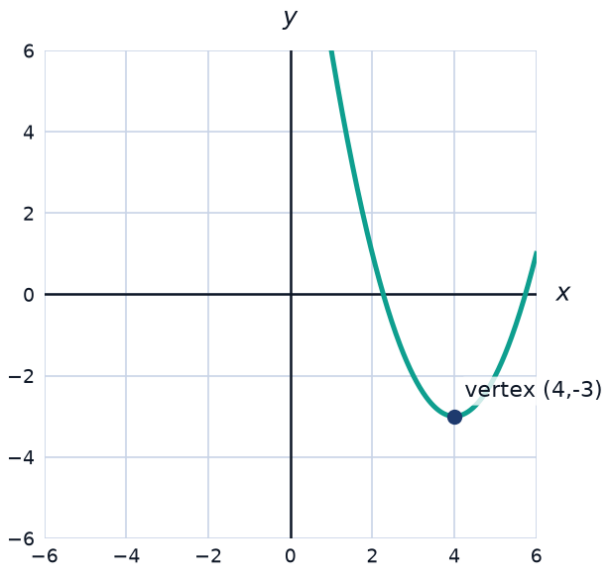
c. Reflecting over the x-axis

d. Shifting 4 units left and 5 units down

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34 **[Intermediate]** • **Target: Evaluating and describing**

The graph of $g(x) = f(x - 4) - 3$ is obtained from $f(x)$ by:



a. Reflecting over the x-axis

b. Shifting 4 units right and 3 units down

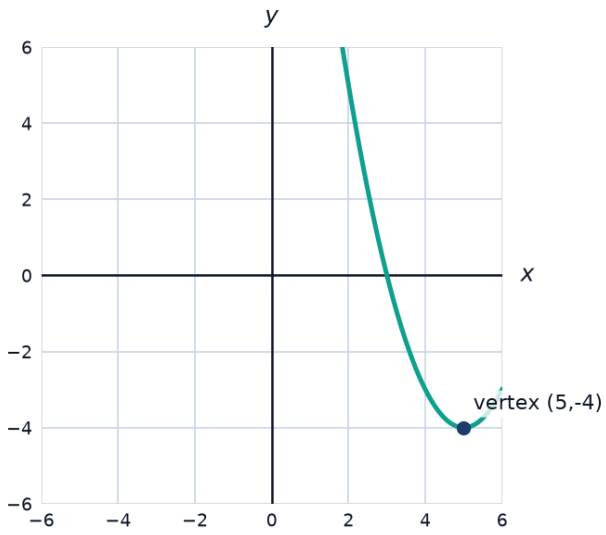
c. Shifting 4 units left and 3 units down

d. Shifting 4 units left and 3 units up

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35 **[Intermediate]** • **Target: Identifying from graphs and applications**

The graph of $g(x) = f(x - 5) - 4$ is obtained from $f(x)$ by:



a. Shifting 5 units left and 4 units up

b. Shifting 5 units right and 4 units down

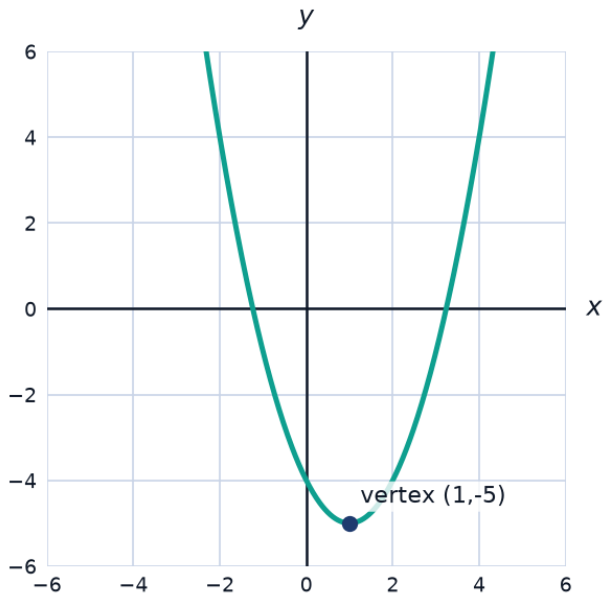
c. Reflecting over the x-axis

d. Shifting 5 units left and 4 units down

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36 [Intermediate] • Target: Compound interest

The graph of $g(x) = f(x - 1) - 5$ is obtained from $f(x)$ by:



a. Shifting 1 units right and 5 units down

b. Shifting 1 units left and 5 units down

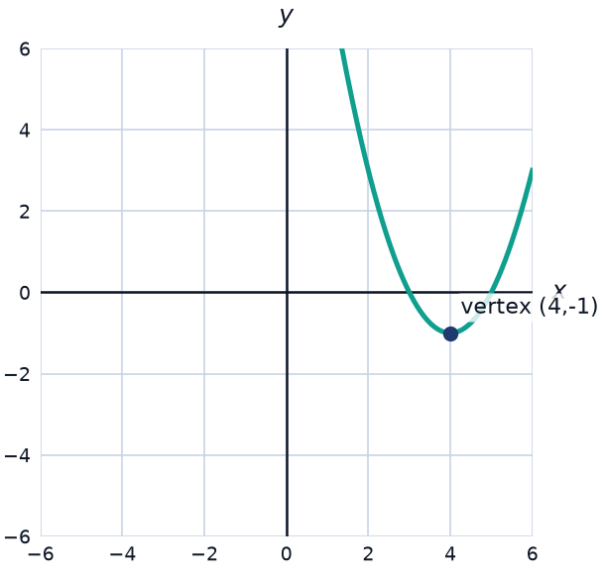
c. Reflecting over the x-axis

d. Shifting 1 units left and 5 units up

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37 **[Intermediate]** • **Target: Comparing exponential growth rates**

The graph of $g(x) = f(x - 4) - 1$ is obtained from $f(x)$ by:



a. Reflecting over the x-axis

b. Shifting 4 units left and 1 units up

c. Shifting 4 units left and 1 units down

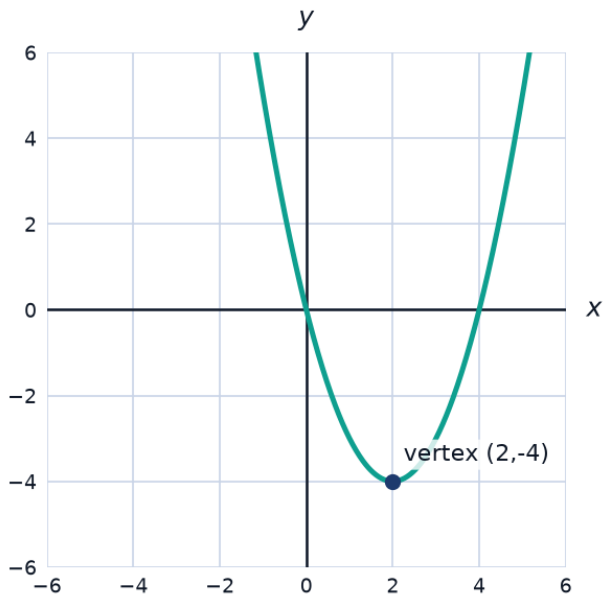
d. Shifting 4 units right and 1 units down

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Comparing growth and decay graphically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

The graph of $g(x) = f(x - 2) - 4$ is obtained from $f(x)$ by:



a. Reflecting over the x-axis

b. Shifting 2 units left and 4 units up

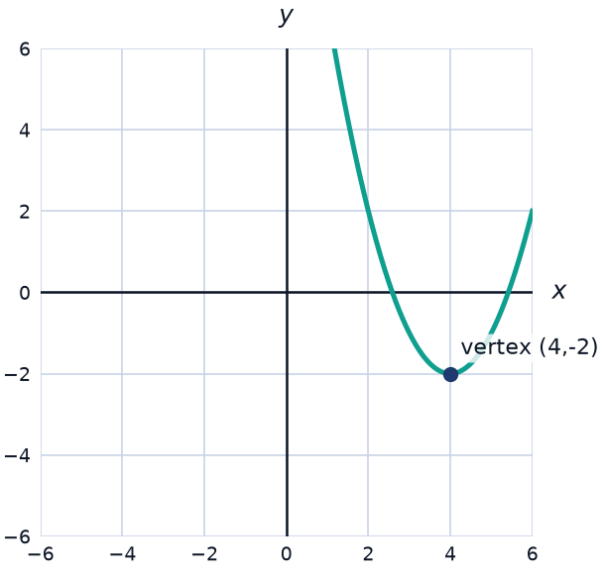
c. Shifting 2 units left and 4 units down

d. Shifting 2 units right and 4 units down

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39 [Challenging] • Target: The natural exponential base e • Cross-Lesson: 3.03 Sum and Difference Formulas

The graph of $g(x) = f(x - 4) - 2$ is obtained from $f(x)$ by:



a. Shifting 4 units left and 2 units up

b. Shifting 4 units left and 2 units down

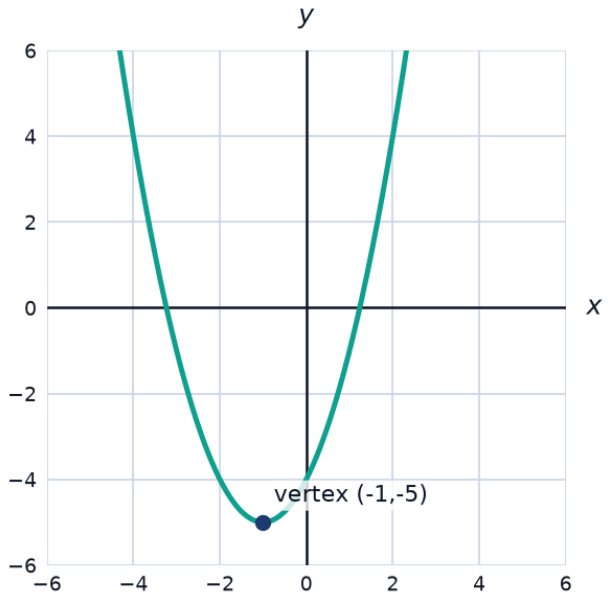
c. Reflecting over the x-axis

d. Shifting 4 units right and 2 units down

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40 [Challenging] • Target: Piecewise-exponential applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:



a. Shifting 1 units left and 5 units down

b. Shifting 1 units right and 5 units down

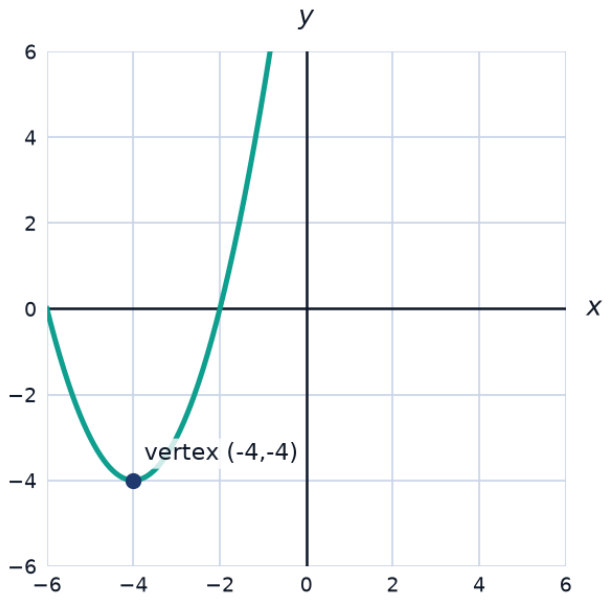
c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

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41 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 4.01 Circles and Parabolas

The graph of $g(x) = f(x + 4) - 4$ is obtained from $f(x)$ by:



a. Shifting 4 units right and 4 units down

b. Shifting 4 units left and 4 units down

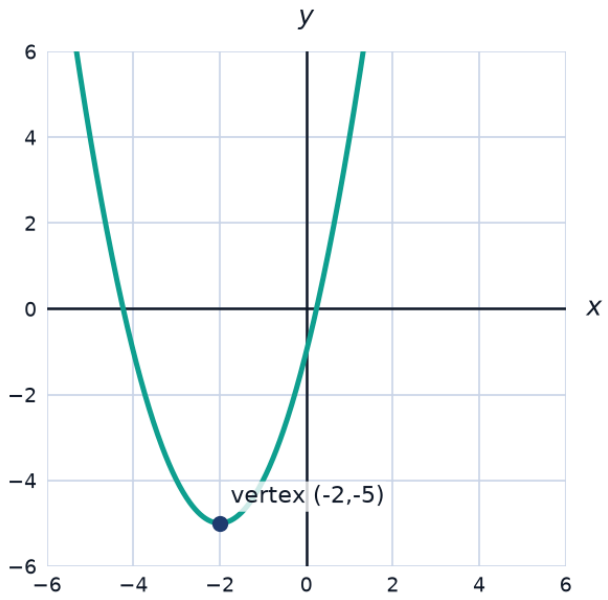
c. Shifting 4 units right and 4 units up

d. Reflecting over the x-axis

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42 **[Challenging]** • **Target: Comparing exponential functions numerically** • **Cross-Lesson: 4.02 Ellipses**

The graph of $g(x) = f(x + 2) - 5$ is obtained from $f(x)$ by:



a. Shifting 2 units right and 5 units down

b. Shifting 2 units right and 5 units up

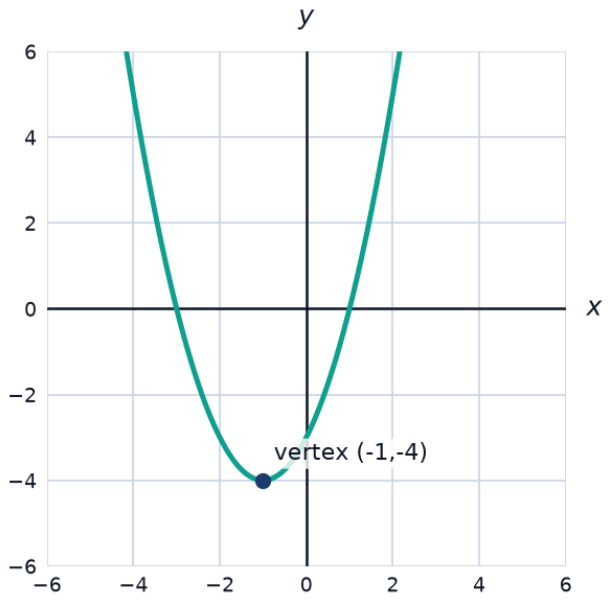
c. Reflecting over the x-axis

d. Shifting 2 units left and 5 units down

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43 [Challenging] • Target: Base e and Exponential Growth/Decay Curves • Cross-Lesson: 4.03 Hyperbolas

The graph of $g(x) = f(x + 1) - 4$ is obtained from $f(x)$ by:



a. Shifting 1 units left and 4 units down

b. Shifting 1 units right and 4 units up

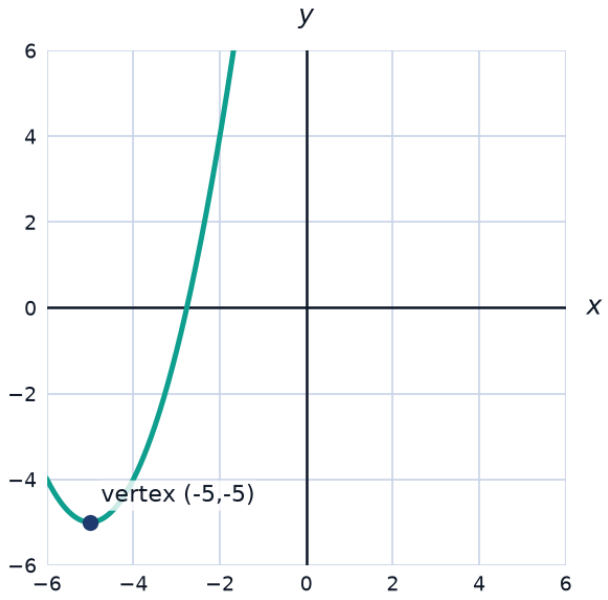
c. Shifting 1 units right and 4 units down

d. Reflecting over the x-axis

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44 **[Challenging]** • **Target: Transformations of Exponential Functions** • **Cross-Lesson: 5.01 Vectors: Components and Operations**

The graph of $g(x) = f(x + 5) - 5$ is obtained from $f(x)$ by:



a. Shifting 5 units right and 5 units up

b. Shifting 5 units left and 5 units down

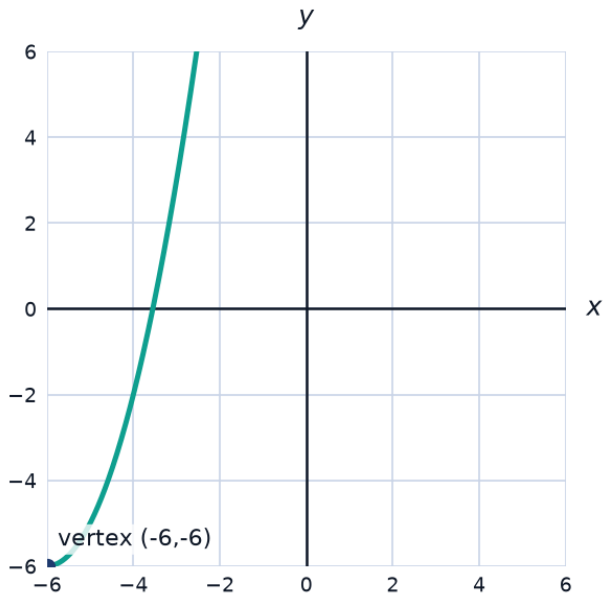
c. Reflecting over the x-axis

d. Shifting 5 units right and 5 units down

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45 [Challenging] • Target: Compound Interest Formulas • Cross-Lesson: 5.02 The Dot Product

The graph of $g(x) = f(x + 6) - 6$ is obtained from $f(x)$ by:



a. Reflecting over the x-axis

b. Shifting 6 units left and 6 units down

c. Shifting 6 units right and 6 units up

d. Shifting 6 units right and 6 units down

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